A Measured Datasheet for the iPhone 16 Pro Max Main Camera: Geometry, Resolution, Noise, and Color

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Repository: https://github.com/Jon-Knapp/EE513
Resume line: _____

***Abstract*—I characterized the main camera of an iPhone 16 Pro Max from its Bayer raw files, and give every figure with an uncertainty and the method that produced it. A 25-view checkerboard calibration gives a focal length of 2901.8 ± 4.6 px with 0.86 px RMS reprojection error. A slanted-edge measurement gives an MTF50 of 0.403 ± 0.032 cycles per green-plane pixel at the centre, falling to 68% of that 32% of the way to the corner. A difference-of-pairs photon transfer analysis at ISO 50 gives a conversion gain of 0.246 ± 0.021 DN/e^-, a read noise of 6.9 ± 0.6 e^- and a dynamic range of 11.04 ± 0.06 stops. A least-squares color correction matrix reduces the mean CIEDE2000 error on a ColorChecker from 12.35 to 3.06 ± 0.04. Each result is compared with the manufacturer's figures where they exist.**

***Index Terms*—Camera calibration, color correction, dynamic range, image sensor noise, modulation transfer function, photon transfer curve.**

I. Introduction

Megapixels and an f-number say little about image quality. This report measures what does, for one camera: the main (1×) rear camera of an iPhone 16 Pro Max. All captures are Bayer DNGs, and the `lens_id` check from the assignment passes on every series (Table I).

The sensor is a 48 MP quad-pixel design, and its 12 MP raw output is 2 × 2 binned. Every figure here is therefore per *output* pixel: a 2.44 µm pitch [1], [2], not the 1.22 µm photodiode.

Uncertainties use four conventions. "s.d." is the standard deviation over repeated captures. "Jackknife" is the leave-one-view-out standard error. "Fit" is the 1σ from a fit's covariance. A "bound" is a limit stated with its argument.

Table I
Device Under Test

Item	Value
Camera (EXIF)	iPhone 16 Pro Max back camera 6.765mm f/1.78
Raw format	Bayer DNG, 4032 × 3024, BGGR, 12-bit; black 528 DN, white 4095 DN
Capture app	Halide Mark III - Pro Camera v3.3.3
Same-camera check	`lens_id` identical over 25 calibration, 6 slanted-edge, 20 flat-field, 23 dark and 12 color captures

II. Measured Datasheet

Table II summarizes the results; Sections III to VII give the method behind each one. Three of them can be compared with the manufacturer.

Pixel pitch. Apple's binned pitch is 2.44 µm [1]. The EXIF crop-factor route gives 2.422 µm (−0.7%), and the calibration route gives 2.331 µm (−4.5%).

Focal length. The calibrated value, 7.03 ± 0.15 mm, is 3.9% above the EXIF 6.765 mm. I attribute most of the gap to focus breathing. The calibration board was 234 to 365 mm away (from the extrinsics), and focusing a 6.765 mm thin lens that close moves it 1.9% to 3.0% further from the sensor.

Noise model. The DNG's NoiseProfile tag [8] is the vendor's own model; it is compared with my measurements in Section V.

TABLE II
MEASURED DATASHEET

	Quantity	Value	Uncertainty from
Geometry	Focal length	2901.8 ± 4.6 px	jackknife, 25 views, 5-coef. model
	Focal length	7.03 ± 0.15 mm	f_px × 2.422 µm; bound from 24 mm-equiv. rounding
	Principal point (x, y)	2026.1 ± 4.3, 1510.0 ± 4.4 px	jackknife
	Radial k1, k2, k3	0.232 ± 0.008, -0.946 ± 0.038, 1.07 ± 0.063	jackknife
	Tangential p1, p2	0.001 ± 0.00051, -0.00075 ± 0.00062	jackknife
	RMS reprojection error	0.863 ± 0.029 px	jackknife
Resolution	MTF50, centre	0.403 ± 0.032 cy/green px	s.d., 6 captures; Nyquist = 0.5
	MTF at green Nyquist, centre	0.404 ± 0.032 (predicted 0.785)	s.d., 6 captures
	MTF50, corner squares (32% field)	0.275 ± 0.022 cy/green px	s.d., 6 captures; not the frame corners
	Corner-square / centre MTF50	0.681 ± 0.010	s.d. of the per-capture ratio
Noise (ISO 50)	Conversion gain K	0.246 ± 0.021 DN/e^-	fit ± fit-range systematic, in quadrature
	Read noise	6.9 ± 0.6 e^- (1.694 ± 0.002 DN)	dark-frame pairs; gain uncertainty propagated
	Full well (clip level)	14,500 ± 1,200 e^-	3567 DN / K
	Shot/read-noise knee	47 ± 8 e^-	σr^2
	Dark current	< 0.6 e^-/s (95%)	bound, 23 capped frames, 1/30 to 1 s
	Black level	527.4 ± 0.1 DN (file: 528)	spread between channels
Dynamic range	Full scale / read noise	11.04 ± 0.06 stops	bound from the effective white level
Color (LED)	Mean ΔE00 before CCM	12.35 ± 0.08	s.d. over 3 charts
	Mean ΔE00 after CCM	3.06 ± 0.04	s.d. over 3 charts
	Worst patch after CCM	8.89 ± 0.26 (15, red)	s.d. over 3 charts

III. GEOMETRY: INTRINSICS AND DISTORTION

A. Calibration

I calibrated with `cv2.calibrateCamera` [3], [4] on 25 views of a 9 × 6 board with 20.0 mm squares; all 25 were detected. Eight views deliberately placed the board near the edges and corners of the frame. With four distortion coefficients (k1, k2, p1, p2) the RMS reprojection error is 1.44 px. Adding k3 lowers it to 0.86 ± 0.03 px, so I adopt the five-coefficient model (Table III).

B. Uncertainty

The fit's own 1σ understates the uncertainty by a factor of 3 to 6. For example, the principal point is ±1.3 px from the fit but ±4.3 px from the jackknife. The fit treats all 54 corners of a view as independent, but the errors within a view are correlated, so the jackknife uncertainty is the one reported. No single view drives the spread: leaving any one out moves the principal point by at most 1.8 px.

Board flatness was checked separately. A homography fit to each view's undistorted corners leaves a 0.67 px median residual (0.91 px max), with each view left out of its own calibration. Even if all of that residual were warp, the board would be flat to about 0.2 mm.

The tangential terms are within 2σ of zero. The principal point sits 10.6 px right of and 1.5 px above the geometric centre. The detected corners reach a normalized radius of 0.676, against 0.872 at the frame corner, so the outer 22% of the radius is extrapolated.

C. Undistortion

The undistorted frame is shown in Fig. 1. On the board row furthest from the principal point, the maximum deviation from a straight line falls from 1.35 px to 0.69 px. Over all 150 rows, the median falls from 1.36 px to 0.52 px.

TABLE III
CALIBRATION WITH FOUR AND FIVE DISTORTION COEFFICIENTS (± JACKKNIFE)

Model	f (px)	c_x, c_y (px)	k1	k2	p1	p2	k3	RMS (px)
4 coef.	2914.6 ± 6.2	2030.7 ± 12.3, 1515.4 ± 9.6	0.114 ± 0.012	−0.268 ± 0.031	0.0017 ± 0.0013	−0.0000 ± 0.0014	0 (fixed)	1.44 ± 0.11
5 coef. (adopted)	**2901.8 ± 4.6**	**2026.1 ± 4.3, 1510.0 ± 4.4**	**0.232 ± 0.008**	**−0.946 ± 0.038**	**0.0010 ± 0.0005**	**−0.0007 ± 0.0006**	**1.070 ± 0.063**	**0.86 ± 0.03**

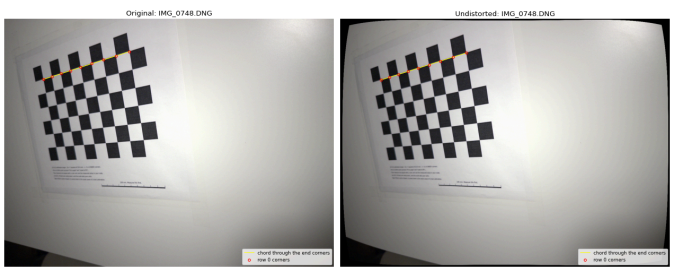


Fig. 1. Original frame (left) and the same frame undistorted with the five-coefficient model (right). Red circles mark the board row furthest from the principal point; the yellow chord joins its end corners.

IV. SPATIAL RESOLUTION

A. Method

I measured the MTF with the slanted-edge method [5] on one black-subtracted green plane of the Bayer mosaic, so no demosaicing interpolation enters the edge. A green-plane pixel spans 2p = 4.84 µm. All MTF values are in cycles per green-plane pixel, where Nyquist is 0.5; halve them for full-mosaic pixels. On a synthetic Gaussian edge with a known answer (σ = 1 px), the code's MTF50 was within −0.12%.

B. Centre

The centre MTF50 is 0.403 ± 0.032 cycles per green-plane pixel, 0.81 of Nyquist (s.d. over 6 captures, each the mean of 4 edges). At Nyquist the MTF is 0.404 ± 0.032. Fig. 2 compares the measured curve with a diffraction (f/1.78, 550 nm) × pixel-aperture prediction, which gave 0.785 at Nyquist. The camera keeps about half the predicted contrast; lens aberrations are the leading suspect, because the loss grows away from the axis.

C. Field Dependence

The target's corner squares lie at 32% to 33% of the half-diagonal (Fig. 3). Their mean MTF50 is 0.275 ± 0.022, and the corner-to-centre ratio is 0.681 ± 0.010. The ratio is far steadier than either value alone, because a change of focus between captures moves all squares together. MTF50 at the frame corners themselves was not measured, and it is `null` in results.json.

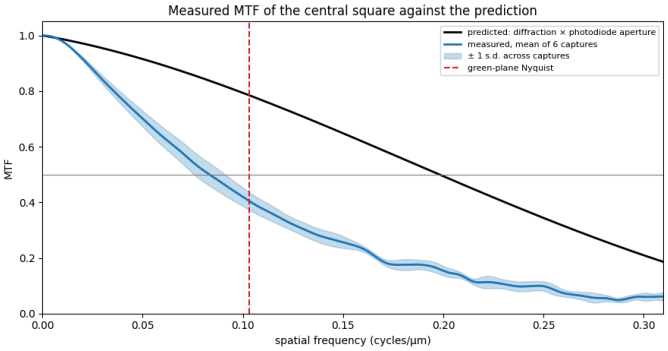


Fig. 2. Measured centre MTF (mean ± 1 s.d. over 6 captures) against the diffraction × photodiode-aperture prediction. The dashed line marks green-plane Nyquist.

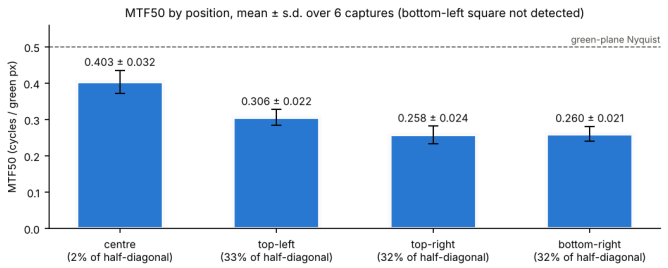


Fig. 3. MTF50 by position on the target (mean ± s.d. over 6 captures). The bottom-left square was not detected.

V. Noise: The Poisson–Gaussian Model

A. Photon Transfer

The flat fields were 10 pairs from 0.9% to 96% of saturation, at ISO 50 and 1/30 to 1 s. The light was an overcast window through 2 to 10 sheets of paper. Statistics come from a 128 × 128 patch at the centre of one green plane, because lens shading darkens the rest of the frame.

For each pair, the mean is the average of the two frames above black, and the variance is ½Var(A − B). I fit $\mathrm{Var} = K\mu + K^2\sigma_r^2$ [6], [7], weighted by 1/variance, through the 9 pairs below 70% of saturation. Before taking σ_r, I removed 1/12 DN² of quantization variance from the intercept.

Single frames are not enough, because a single frame's variance includes fixed-pattern PRNU, which grows as $(p\mu)^2$. Fitting single frames gave K 14.3% higher (0.281 DN/e⁻), with p = 0.61%. The difference of a pair cancels that pattern.

B. Results

The gain is K = 0.246 ± 0.021 DN/e⁻. That uncertainty combines the fit's ±0.014 with a ±0.015 systematic: K is 0.218, 0.229 and 0.246 when the fit stops at 15%, 30% and 70% of saturation.

The fit's intercept constrains the read noise poorly, at 4.2 ± 2.9 e⁻. Measured directly from 11 dark-frame pairs, it is 1.694 ± 0.002 DN, or 6.9 ± 0.6 e⁻.

The two regimes meet at $\mu = \sigma_r^2 = 47 \pm 8$ e⁻. The darkest pair (135 e⁻) is 2.9× above that knee, so the data show the start of the bend but not the read-noise regime itself (Fig. 4).

C. Vendor NoiseProfile

The NoiseProfile tag [8] gives $S = 8.858 \times 10^{-5}$ and $O = 9.874 \times 10^{-7}$. Scaled by 3567 DN, these correspond to K = 0.316 DN/e⁻ and σ_r = 11.2 e⁻, which are 22% and 52% above my values. The profile matches my measured variance at 54% of saturation but overstates it toward black, by 2.3× at the darkest pair.

D. Dark Current

I fitted 23 lens-capped frames at four exposure times from 1/30 to 1 s. The four Bayer channels disagree beyond their errors, and the dark level jumps between frames at a fixed exposure by as much as the whole trend. The fitted slope is therefore not a measurement. I report a one-sided 95% upper bound of < 0.6 e⁻/s, which adds less than 1.3% to the read-noise variance at the app's 1 s maximum.

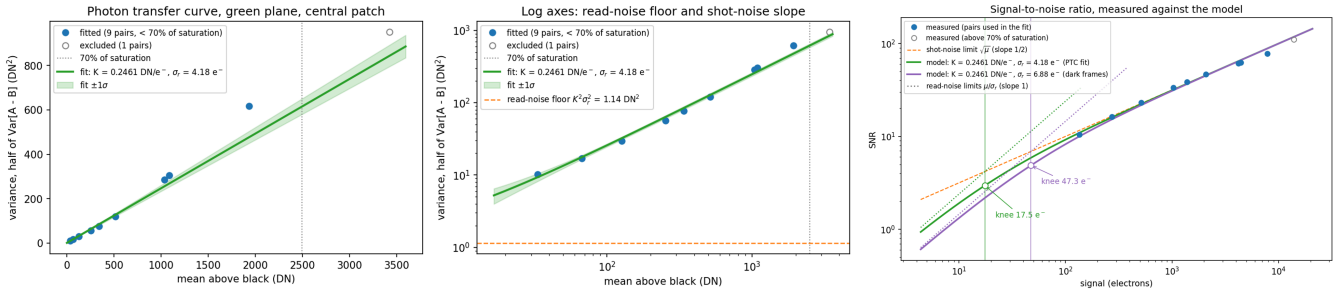


Fig. 4. Photon transfer curve from the central green-plane patch on linear and logarithmic axes (left), and measured SNR against the model with the knee marked (right).

VI. Dynamic Range

The dynamic range is DR = $\log_2$(3567 DN / 1.694 DN) = 11.04 ± 0.06 stops. The top of the range is white level minus black level. The bottom is the signal equal to the read noise, which is SNR = 1 from read noise alone. In DN, this ratio does not depend on the gain.

The ± is a bound. The green sites begin to saturate near 3950 DN rather than 4095 DN, and that ceiling would give 10.98 stops. For comparison, the EMVA 1288 criterion [7] (SNR = 1 including shot noise) gives 10.9 stops, and SNR = 10 gives 6.7 stops.

VII. Color Accuracy

A. Method

I photographed three ColorChecker Classic Mini charts under an LED lamp at ISO 400 and 1/60 s; no patch clipped. The raw data were normalized to the measured 3940 DN saturation level and reduced to quarter-resolution RGB without demosaicing. Lens shading was corrected per channel from a white sheet. Each patch value is the mean of its middle half.

White balance came from patch 22. A 3 × 3 matrix to XYZ was then fitted by least squares over all 24 patches.

Errors are CIEDE2000 [9], against colour-science's *ColorChecker24 – After November 2014* set [10]. I chose that set because X-Rite reformulated the chart in November 2014.

B. Results

Before the matrix, mean ΔE00 is 12.35 ± 0.08. After it, the mean is 3.06 ± 0.04, and the worst patch is 15 (red) at 8.89 ± 0.26 (Fig. 5). The uncertainties are the s.d. over three charts, which agree to about 1%.

The residual remains because the sensor fails the Luther condition: its spectral sensitivities are not a linear combination of the color-matching functions. Carried unchanged to other lights, the same matrix gives 5.79 under a blackbody lamp and 6.94 under daylight.

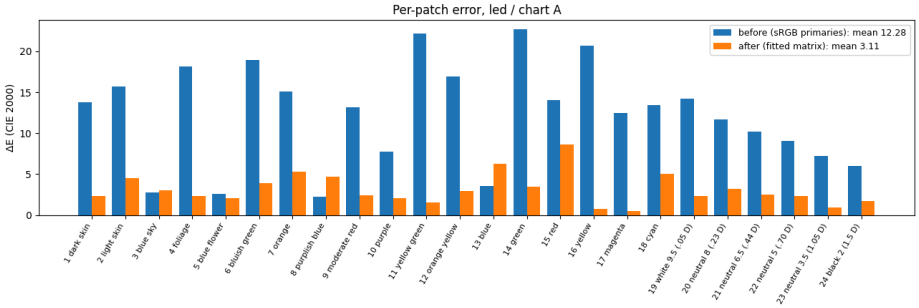


Fig. 5. CIEDE2000 error per patch before and after the fitted matrix (LED, chart A).

References

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Decisions and Verification Statement

What I decided, and what I rejected.

- **Five distortion coefficients, not four.** The board positions near the corners pushed the four-coefficient RMS to 1.44 px. With k3 it is 0.86 px.
- **The EXIF-route pixel pitch (2.42 µm), not the calibration route (2.33 µm).** The calibration's focal length is inflated by close focus.
- **MTF on one green plane** of the raw mosaic, never on demosaiced data.
- **Read noise from dark-frame pairs, not the PTC intercept.** The intercept is an extrapolation with ±69% uncertainty.
- **Dark current reported as an upper bound, not the 0.33 e^-/s fit.**
- **Saturation at 3940 DN, not the declared 4095.** The green sites pile up from about 3950 DN, so I did not trust the declared white level. That excluded seven blackbody patches and made the LED round the primary color round.
- **Frame-corner MTF left `null`** in results.json rather than standing in the 32%-field values.

AI use. I used Claude (Anthropic) to write most of the analysis code in the four homework notebooks, to diagnose problems in my captures, to explain concepts, and to draft written answers, which I reviewed and edited. Claude also assembled this report and results.json from those notebooks, and wrote `project1_checks.py`, which adds the jackknife calibration uncertainties and the corner-square MTF.

How I checked the results.

- **Known answers before real data.**
  - The slanted-edge code recovered a synthetic σ = 1 px Gaussian edge's MTF50 to −0.12%.
  - The PTC code, on the lecture 5 simulator, recovered K to −0.04% ± 0.21% and σr to +0.30% ± 0.70% over 20 independent runs.
  - The sRGB encode/decode passes a round trip to 1e-12.
- **Failure modes I found and fixed.**
  - My first read-noise fit on the simulator was **+5.2% (8σ) high**. The cause was the 1/12 DN^2 that rounding adds to every variance. With it subtracted, the bias is gone.
  - **A flatness test that mixed two effects.** In Homework 2, a homography fit to the raw corners left a 2.22 px median residual, and I took that as evidence the board was not flat. Re-run on undistorted corners, with each view left out of its own calibration, the residual is 0.67 px median and 0.91 px max. 86% of the raw residual pointed along the radius, so it was lens distortion, not warp; the board is flat to about 0.2 mm. The jackknife uncertainties stand, but their cause is correlated corner errors within each view, which I have not traced.
  - A fixed 11 px corner-refinement window failed on defocused close-range boards. I made it scale with the square size.
  - The slanted-edge target took four capture sessions before repeated photos agreed. The earlier sessions showed close-range defocus and off-centre framing.
- **Independent cross-checks.**
  - Pixel pitch two ways, against Apple's 2.44 µm.
  - Read noise from the fit against the dark frames (1.3σ apart).
  - Single-frame against pair gain. The +14.3% matches the +14.2% the simulator gives for a known PRNU.
  - Color across three physical charts (3.03 to 3.11).
  - Re-running the Homework 2 functions inside `project1_checks.py` reproduced the centre MTF50 exactly (0.4035).
- **One thing the checks caught here.** The calibration's own 1σ values were 3 to 6× smaller than the leave-one-view-out jackknife (principal point ±1.3 px against ±4.3 px). I report the jackknife.

What I am still unsure about.

- **Frame-corner MTF50 is not measured.** The fourth corner square was not detected either.
- **Variance/mean rises with signal** (0.22 to 0.32). I cannot tell sensor non-linearity from processing in Apple's raw path, so the gain carries a ±0.015 DN/e^- systematic.
- **The vendor NoiseProfile disagrees with my read noise by 2×,** and I do not know how Apple derives it.
- **The checkerboard square size.** I read 20 mm with a ruler (±0.5 mm). The slanted-edge sheet's 100 mm scale bar measured 95.80 mm with calipers, so the checkerboard may also be printed below 100%. This does not change f_px, the principal point or the distortion, which are scale-free, but it scales the board distances, by about 4%.
- **The focus-breathing account** is a thin-lens estimate (+1.9% to +3.0%) of a 3.9 ± 2.1% gap. It is consistent, not proven.
- **I did not verify the color charts' production date** against the post-2014 reference set.